



(Credit: www.sigmaaldrich.com)

out with a long-term aim to eliminate the demand-availability gap through 3D bioprinting. In this type of additive manufacturing, biomaterials like cells and other biocompatible materials are combined to form bioinks. After testing the suitability of bioinks, these are then printed to generate bodily structures such as tissues and organs that imitate natural ones.

The company launched its first product, the bioprinter Trivima, meaning three-dimensional in Sanskrit, at

Bengaluru Tech Summit in 2018. The design of most machine components was done using Autocad Fusion 360. The precise, customisable and lightweight 3D bioprinter specifically caters to the research requirements of individuals in the field of 3D bioprinting. Some examples include the development of tissue disease models

and 3D printed scaffolds, stem cell research, and the like. The possibility of varying print speed, temperatures, nozzle diameters, and parameters in various mechanisms helps researchers in better investigation of their topic of interest, which is not present in many other bioprinters in the market.

The startup has already expanded its business by setting up offices in Germany in 2018 and it looks forward to more such opportunities. It has also developed Innoskin human epidermis

(HE) using its patented 3D bioprinting technology, which is the first in the series of Innoskin products. Innoskin FT, which is the other variant, consists of dermal and epidermal components of the skin.

The challenge was to use the right materials that could combine to form human skin. The product can replace animal testing for cosmetic, chemical, agrochemical, and pharmaceutical sectors working on skin-related applications, which is the current standard method in most places. The company plans to offer it later on for clinical applications where 3D bioprinted tissue can be used as a skin graft to treat skin injuries, burn victims and other skin related damages.

Fully functional bioprinted organs will need to clear the necessary guidelines before being used as a transplant. But once this becomes successful, it will have a huge impact on the lives of common people by increasing availability and affordability.

—Deepshikha Shukla

4 CONTACT-FREE IoT BIOSENSORS FOR PATIENT MONITORING

It is often seen that late diagnosis of many diseases leads to loss of patients' lives. In most cases, monitoring key parameters like respiration and heart rates, and hence early prevention as a response to warning signs can lead to better patient care, reduced hospital stays, and lower costs.

This led to the establishment of Vitalet in Israel in 2016 with an aim to enable continuous preventive monitoring of vital signs for predicting future health risks for every person. In order to achieve this, the company designed a comprehensive Internet of Things (IoT) patient monitoring platform with the same name. The integrated platform uses small, lightweight, contact-free IoT biosensors along with artificial intelligence (AI) software to ensure accuracy.

The mobile biosensor continuously monitors heart rate, respiration rate, and body

motion. It can be easily attached under the patient's bed or wheelchair without any discomfort as compared to already available pressure mats in the market. The biosensor is wirelessly connected to the integrated care management system. Data including reports and graphical data analysis is sent via Wi-Fi to the cloud based

machine-learning platform. It also alerts medical staff when a patient requires assistance. The alerts can be prioritised by the doctors so that the most important cases are not neglected while avoiding alarm fatigue. This ensures close monitoring without the need for one-to-one nursing care all the time.

The advanced cloud based alerting system provides real-time actionable information from anywhere and a customisable monitoring dashboard identifies the risk of falling signals to avoid falls before they happen, and tracks bed-time and turn movements for ulcer prevention. Monitoring is also done to send irregular breathing alerts and heart incident alerts, improve sleep quality, and predict epileptic seizures, among others.

The device is waterproof with no physical wear, and connects via Bluetooth or Wi-Fi

